StockPulse — Master Requirements Document

This is the single source of truth for the project. Every confirmed decision goes here. Nothing gets built or changed without it being reflected in this file first.

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1. The Problem
- SMB owners running 2-3 online stores + a physical warehouse have no single view of stock
- Price creep from suppliers goes unnoticed until it's already cost money
- Owners don't want another dashboard to log into — they live in WhatsApp

2. ICP (Target Customer)
- SMB owners in Pakistan running multiple sales channels
- Currently tracking stock manually or on disconnected spreadsheets
- Have 1-3 staff needing some level of access
- Price-sensitive — can't afford global enterprise tools

3. Core Features (confirmed)
- One dashboard pulling stock/sales data from every store + warehouse
- Daily automatic report sent to WhatsApp
- AI flags purchase-price anomalies the moment a new price is entered
- Two-way WhatsApp Q&A (ask "biscuits ka stock kitna hai?", get an answer)
- Role-based access tied to WhatsApp number — no separate logins
- **Price comparison — two separate functions, precisely:**
  - **(a) Instant, event-driven:** every time a new purchase price is entered, compare it to that product's last recorded purchase price — not tied to any calendar date, fires whenever stock is actually added. Significant changes trigger an immediate WhatsApp alert ("📈 Biscuits: PKR 145, up 12% from last purchase")
  - **(b) Monthly summary chart:** generated automatically at the end of each calendar month, comparing that month's purchase prices per product against the previous month's — sent as one WhatsApp image. Runs independently of (a), not a replacement for it.
- **Language selection:** chosen once at setup (English / Urdu / Roman Urdu), applies to every report, alert, and Q&A reply after that
- **Daily backups (corrected architecture):** live data lives in a real persistent hosted database (not a local SQLite file on Vercel, which cannot be trusted to persist between deployments). A full snapshot is exported automatically every day, stamped with date and time, to a separate storage location. A **Backup History** screen (in Settings) shows the full list of past backups, not just the most recent one.
- **Multi-tenant architecture:** one shared "skeleton" (core app/code), but every shop owner's data is completely private and isolated — Shop Owner 1's grocery items and Shop Owner 2's clothing sizes never mix or see each other. Each owner sets their own categories/products inside their own private space.

4. Why This Approach (differentiation)
- WhatsApp-native, not just WhatsApp-notified — two-way interaction, not just a push report

- Proactive AI price alerts, not something the owner has to notice themselves
- Built for Pakistan specifically — PKR pricing, Urdu/Roman Urdu, local payment methods — not a translated global tool

5. Competitive Landscape
- Zoho Inventory, Extensiv/Brightpearl — solid dashboards, but enterprise-priced (PKR 5,600-84,000+/month), English-only, not WhatsApp-native
- Vyapar, Syncli — WhatsApp-integrated, but built for India's GST/UPI ecosystem, not Pakistan's
- Gap: nobody combines WhatsApp-first interaction + proactive AI price alerts + Pakistan-specific setup in one product

6. Pricing Model
- Starter tier (1 store/warehouse): **PKR 2,000/month**
- Growth tier (2-3 stores + staff access): **PKR 3,500/month**
- Cost per customer (Meta WhatsApp fees): ~PKR 200/month
- Shared BSP platform fee: ~PKR 14,000/month (spread across all customers — shrinks per-customer as you scale)
- **Note:** pricing needs validation with real shop owners before being treated as final

7. Investment & Revenue Projections
To build/run the pilot (rough, one-time + first couple months):
- Cursor subscription: ~PKR 5,600/month
- Domain: ~PKR 3,000-6,000/year
- Hosting: ~PKR 2,000-5,000/month
- WhatsApp costs during pilot: **PKR 0** (using Twilio free sandbox)
- Total to get pilot running: **~PKR 10,000-15,000**

Once live with paying customers (monthly, ongoing):
- Fixed cost floor: ~PKR 20,000-25,000/month (BSP + hosting)
- Plus ~PKR 200/customer/month in messaging fees

Projected profit (estimates only, not guarantees):

Customers	Revenue/mo	Costs/mo	Profit/mo
20	PKR 55,000	~PKR 23,000	~PKR 32,000
50	PKR 137,500	~PKR 38,000	~PKR 99,500
100	PKR 275,000	~PKR 65,000	~PKR 210,000

8. WhatsApp BSP Choice
- **Pilot phase:** Twilio — free sandbox testing, pay-as-you-go, fits a developer-led build with zero upfront WhatsApp cost
- **Post-pilot / going live in Pakistan:** switch to WeTarseel — PKR-native billing, local support, ~PKR 14,000/month platform fee
- Important: from **October 1, 2026**, WhatsApp Service message replies in Pakistan stop being free ($0.015/message) — factor this into live costs, not pilot costs (sandbox testing isn't affected)

9. Demo/Pilot Build Scope
In scope for the demo (5-10 real shop owners testing it):

- Manual stock/price entry for one store/warehouse per owner
- Daily WhatsApp report, auto-sent
- Price-anomaly alert on new price entry
- Basic WhatsApp Q&A (can be semi-manual behind the scenes at first)
- Simple database (Google Sheet or SQLite) — not a big production DB
- Twilio free sandbox for WhatsApp testing

Explicitly OUT of scope for the demo (save for later):
- Multi-channel API integrations (real Shopify/store syncing)
- Voice-note stock entry
- Supplier comparison
- Shrinkage/mismatch detection

9B. "Try With Sample Data" — Architecture Clarification
This is NOT a separate demo app. It is a feature inside the one real app, inside each owner's own private, isolated account:
- When an owner signs up, a real account is created — private, backed up, fully connected to the real system, same as any other account
- "Try with sample data first" (offered on the Setup screen) simply pre-fills that owner's own account with a few dummy products, so they can explore the real app before trusting it with their actual stock
- Real backup, storage, WhatsApp connection, and staff access are active the whole time — none of it is a stripped-down or throwaway version
- Once the owner is ready, sample products are either cleared or clearly separated from their real entries (see Prompt 5, "no mixing" rule) — the pilot owners' real usage and feedback carries forward, it is never a disposable trial that gets wiped

10. Visual Design — Confirmed
- **Palette: Cream & Navy (lightened)** — warm off-white/cream background (#FAF6EE), lightened navy primary (#2D3B5E), lightened but still legible gold accent (#D9A947)
- Feel: professional, calm, trustworthy — not a "startup" look
- White cards on the cream background for content, soft rounded corners, no harsh shadows
- **Language: full dropdown**, not just a few buttons — English, Urdu, Chinese, German, Arabic, and more. Defaults to showing just the selected language (e.g. "English ▾") collapsed; opens into a clean list on click with a soft, smooth light hover/highlight effect following the cursor — not an instant hard color flash. Chosen once at setup, changeable later in Settings
- This is now the locked direction for every screen

11. Dashboard — Confirmed Layout
- Top bar: business name, language dropdown, "Add product" button
- Status row: last backup timestamp, number of connected stores
- Four stat tiles: total stock value, **different items (clickable — see Item Breakdown below)**, price alerts count, **low stock (clickable — opens filtered list of exactly which items are low)**
- Recent price changes list, with up/down indicators and % change
- Link to full price history/comparison view

12. Item Breakdown Screen (opens from "Different items" tile)

- Back arrow + "All items (count)" header
- One row per product: name, total quantity, and a **warehouse / shop / online store split** shown as badges (e.g. "Warehouse: 30" / "Shop: 12" / "Online: 6") — only shown for locations that actually hold stock for that item. This is how an owner sees what's in the storeroom vs. the shop floor vs. listed online, per item

13. Add Product Screen
- Fields: product name, category (owner-typed, not hardcoded), location (Warehouse / Shop / Online Store dropdown), quantity, purchase price, optional variant/size field
- **Optional photo:** a small "Want to add a photo?" button/link below the main fields, not a required field. Clicking it reveals a photo upload control right there (inline, not a new screen), attached to that same product's name. If the owner never clicks it, the form works exactly as before — this must never feel required or block saving the product
- **Microphone/voice input:** a microphone button next to the input fields so someone who can't comfortably type/read can speak the product name, quantity, etc. instead. This matters directly for staff who may not be comfortable typing in English or at all.
- Same card/input style as the rest of the app

13B. Record a Sale / Stock Deduction (new — closes a gap in the original spec)
- Each location (Warehouse, Shop, Online Store) holds its own quantity for a product — this already exists from the Item Breakdown screen
- **New requirement:** there must be a way to record that stock has sold or moved out from a specific location (e.g. "sold 5 boxes from Shop"), which decreases that location's quantity
- The **total stock quantity automatically recalculates** as the sum of Warehouse + Shop + Online Store — the owner should never have to do this math themselves
- This applies whether the deduction is entered manually (pilot phase) or eventually synced automatically from a connected online store later

14. Price History Screen
- Per-product view, opened from the dashboard's price change list or item breakdown
- Simple bar comparison: last few months plus "this month" highlighted in gold
- This month's price and % change vs last month shown as clear stat rows below the chart

15. Owners & Staff Access Screen (revised — major access model change)

This screen now manages two genuinely different categories of people, not one list with permission levels:

Owners
- Full access to everything — the entire web dashboard, every screen, every chart, every report
- Logs in via WhatsApp OTP (see Section 20 — Login Security)

- Daily reports, price-comparison charts, and any PDFs are sent **only** to owner WhatsApp numbers — never to staff
- **Multiple owners supported.** When adding a second (or further) owner, the existing owner chooses that new owner's access type from a dropdown — this is a configurable setting, not a fixed rule, because the exact model (equal power vs. restricted co-owner) is still being decided and may differ per business:
  - "Equal owner" — full control, including the ability to add/remove other owners
  - "Co-owner" — full dashboard/report access, but cannot remove other owners or change critical account settings
  - The system must support both options existing side by side; do not hardcode one model

Staff (corrected — staff DO get real in-app access, just restricted)
- Staff log in the **same way as owners** — WhatsApp OTP (enter number, get code via WhatsApp, enter code). One login system for everyone, no separate password anywhere in the product.
- **After login, what they see is restricted by role, not blocked entirely:**
  - Staff **never** see the main Dashboard/overview screen (stock value, status line, charts, recent price changes) — that stays owner-only
  - Staff **can** reach functional screens needed to do their job — primarily Add Product (with the voice-input/microphone already built in, which directly covers anyone not comfortable typing)
  - "Can add stock" permission → can open Add Product and submit
  - "View only" permission → can view basic stock lists (e.g. a simple read-only product list) but cannot add or edit anything
  - Staff never reach Price History, Settings, or Staff/Owner management — those stay owner-only regardless of permission level
- Staff **never** receive the daily report, price-comparison charts, or any PDF — those remain owner-only
- List of staff added by **name AND WhatsApp number together** — name is not optional
- Each entry shows when they were added and their permission level
- **WhatsApp conversation remains available as an additional, optional channel** (e.g. a staff member could still message a quick stock question), but it is no longer their *only* way to use the product — the app itself is now their primary interface

16. Settings Screen
- Language (same dropdown pattern as elsewhere)
- Business name
- Backup status — last backup time shown directly, with a visible checkmark confirming it succeeded

17. Animation & Interaction Requirements (locked in from the start, not fixed after the fact)
- Every nav link/button pointing to a section scrolls smoothly to that exact heading — not instantly, not slightly off target
- Page transitions feel smooth, not an abrupt reload
- Every button, card, and link has consistent, visible hover feedback across the entire app, defined once in Prompt 1 — this was a repeated

problem on the portfolio project and must not repeat here

18. Standing Accessibility Rule (applies to every future addition)
Before any feature is added to this app going forward, it must be checked against: **can someone who cannot read (in any language) or cannot type actually use this?** If a screen requires typing, it needs a voice/microphone alternative. If a screen requires reading text to recognize a product, it needs a photo alternative where one exists. This is not optional polish — it is core to who this product is for, and should be checked automatically, not only when pointed out.

Fixes from this audit:
- **Setup screen business name field** now needs a microphone option too — not just Add Product
- **Staff Access "add staff" name field** now needs a microphone option too
- **Product photos must actually be shown, not just stored.** Anywhere a product is listed for the owner to recognize (Item Breakdown, Low Stock, Price History product selection), show the photo thumbnail next to the name when one exists — not just on the Add Product screen where it was uploaded. A photo that's never shown again doesn't help anyone recognize anything.
- **No external dashboard logins required for day-to-day verification.** The person building/running this app should never need to log into Turso, Twilio, Vercel, or any other external service just to confirm the app is healthy. A **System Status panel** inside the app itself (Settings) shows database/WhatsApp/backup health in plain language with simple visual indicators (green check / red X) — no jargon, no technical logs, no external links required.

19. Login Security (WhatsApp OTP)
- Owners log in via WhatsApp OTP: enter WhatsApp number → receive a one-time code via WhatsApp → enter the code → logged in
- No passwords anywhere in this product — staying consistent with how staff access already works
- This protects actual data routes, not just the login screen itself — a logged-out visitor must not be able to reach any account's data by navigating directly to a URL
- Session stays active reasonably long on a device, not re-prompting constantly
- Staff never use this login at all — they have no dashboard access to log into in the first place

20B. STANDING REMINDER — Check Before Any Real Deployment/Sale/Demo
Trigger this check any time the person says anything like "it's done," "I'm going to sell it," "deploying it somewhere real," or "showing it to someone":
- ❌ There is currently NO proper BSP (Business Solution Provider) for WhatsApp — only Twilio's development Sandbox, which requires each number to manually join and is not acceptable for real customers
- ❌ There is currently NO payment/trial gating — anyone who completes login can use the full real app for free, indefinitely
- Before confirming anything is ready to sell, deploy for real, or show to an outside party as a finished product, these two items must be

flagged back to the person, every time, even if not asked.

20A. External Requirement Update (from project supervisor) — Tech Stack Realignment
A set of deliverables and tech requirements were handed down (PRD, TRD, App Flow, Design Brief, Schema Document, Implementation Plan). Cross-checked against StockPulse; results below.

Matches, no change needed: Next.js (frontend), Node.js (backend, via Next.js API routes), Vercel (hosting), Git (version control)

Changed: Database moves from the planned Turso/SQLite to **Firebase/Firestore** — see Prompt 9. This is a genuine architecture shift (relational → NoSQL), not a simple swap.

New, not yet addressed in our docs: Logging / error tracking / unit testing — needs to be written into the TRD. DNS/custom domain — a real future production step, not urgent now.

Open/unclear: Email via Resend — StockPulse has no email anywhere in its design (WhatsApp-only by deliberate choice). Not yet clear if this is for internal error alerts (to the builder) or something customer-facing that hasn't been discussed. Needs clarification before building.

Explicitly dropped — do not appear anywhere in StockPulse's docs: "Clipping Module," "Transcribe Module" — these belong to a different project, not StockPulse. "Auth Module - Gmail Enable" — rewritten as "Auth Module - WhatsApp OTP," since StockPulse was deliberately built without Gmail/email/password auth.

Kept, matches directly: "Plans / Paid Plans" phase — maps to our already-identified trial-to-paid payment gating item.

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20. MASTER DISCUSSION LIST — Priority Order, Nothing Skipped

Reordered by actual importance, most critical first. Status noted for each. **BSP stays parked — do not raise unless asked.**

🔴 HIGH PRIORITY

1. **Restore mechanism — decided.** Builder (has direct Turso access) performs restore manually when needed. No self-service "Restore" button inside the app for owners — deliberately, since an accidental tap could overwrite real live data with an old snapshot. A safer, double-confirmation in-app restore flow can be built later if real demand emerges post-pilot. Not building it now.
2. **Verify Prompt 7 & 8 actually work live, end-to-end.** Nothing beyond OTP login has been confirmed working in practice yet:
   - Staff restricted access (Add Product only, no main dashboard)
   - Multi-owner Equal/Co-owner permission enforcement
   - Database persistence on Turso (survives a real redeploy)

   - Daily backup + Backup History (correct date/time, clearly visible)
   - System Status panel accuracy
   - OTP rate-limiting actually blocking excess attempts
3. **Trial-to-paid payment gating — mechanism still undecided.** Agreed necessary before real selling; not yet decided how it should work (time-limited trial? feature-limited free tier? manual per-customer enabling?). Does not block the free pilot, but is the main remaining blocker before selling for real.

🟡 MEDIUM PRIORITY

4. **Multi-owner model specifics** — built as configurable, exact rules/defaults still "I'll tell you," not yet decided
5. **OTP rate-limit specifics** — built, but exact numbers (attempts allowed, cooldown length) not yet discussed
6. **PWA / Add to Home Screen icon (new).** Lets a shop owner add StockPulse to their phone's home screen like a real app icon, no browser bar — purely a frustration-reduction convenience for non-technical users, doesn't change hosting, storage, or backend. Not yet built.
7. **Remaining items from Prompt 1-6 not yet individually re-verified live:** real WhatsApp daily report/alerts/Q&A beyond OTP, stock deduction math across locations, the monthly price chart (3B), voice input on every field (Prompt 4), data isolation between two test businesses

🟢 LOWER PRIORITY

8. **Shopify's integration role** — confirmed Vercel stays as host; Shopify likely becomes a future "online store" channel integration, not urgent, explicitly deprioritized by request for now
9. **Local-only dev login bypass** — agreed safe, not urgent, deferred
10. **Optional extra Vercel link protection** (e.g. Passfort) — current Twilio Sandbox barrier makes this low-urgency for now
11. **Vercel Hobby plan's non-commercial terms** — relevant only once actually selling

⚪ PARKED — do not raise unless asked
12. BSP selection for production WhatsApp
13. Which 5-10 real shop owners will test it, and when
14. Final pricing validation with real owners

Last updated: this conversation. Add to this file — don't replace sections without carrying forward what's already confirmed.